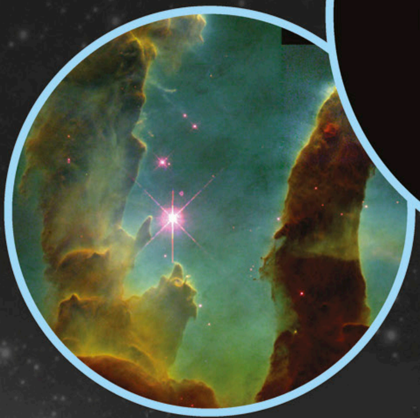
Starry skies

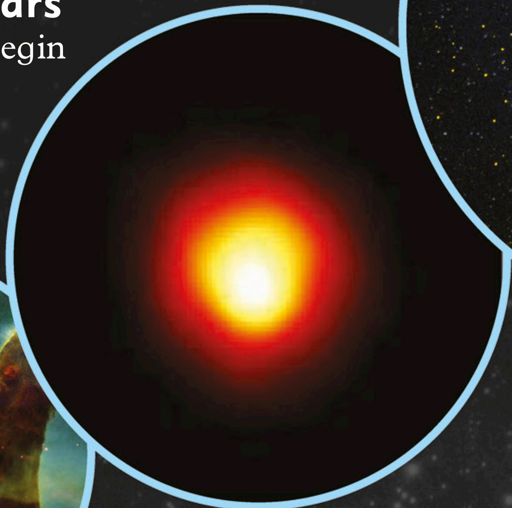
There are many more stars in the universe than there are grains of sand on all the beaches on the Earth. Many are far brighter than our Sun.

The lives of stars

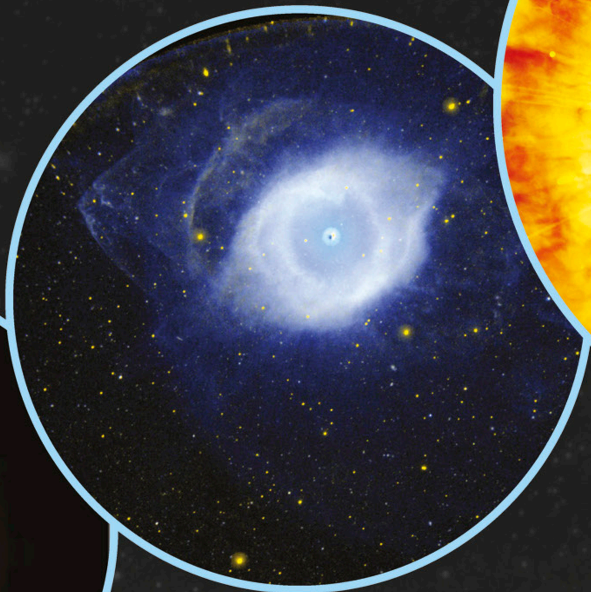
The lives of stars begin inside thick clouds of gas in space, called nebulae.



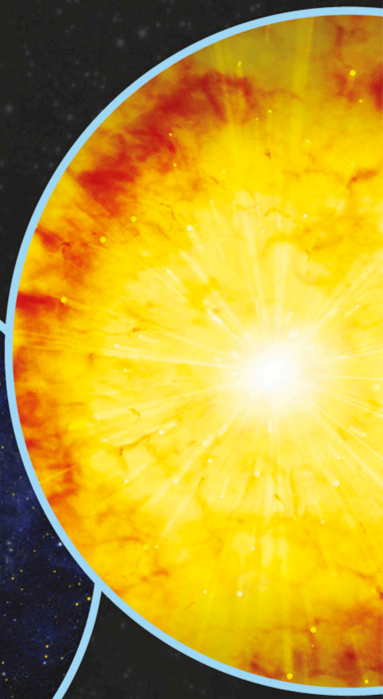
Nebulae
Gravity pulls together little knots of dust and gas inside the nebulae. Each one could become a star, as gravity squeezes it tighter and it becomes hotter.



Red giants
Stars are fueled by the gas hydrogen. They burn until the hydrogen starts to run out. Then they expand, forming a red giant star.



White dwarfs
The outer layers of the star are eventually thrown off into space. The cooling core is left behind. This is called a white dwarf. White dwarfs are no bigger than the Earth.



Supernovae
The most massive stars end their lives in huge supernovae explosions.

Leave a camera shutter open for a few hours on a clear night and you can see the stars leave trails as the Earth rotates.

Stars in motion

The position of the stars seems to change throughout the night. The stars are not really moving, though. It is the Earth that is turning beneath them.

How many stars can you see on a clear, dark night?